Logarithmic algebraic geometry: integral morphisms and log differential forms

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1 Integral and saturated morphisms

The circumstance that fine and fs fiber products do not have the same underlying scheme as the fiber product of schemes can be inconvenient in practice. This is particularly true for log schemes which are not everywhere quasi-coherent. It is therefore useful to have criteria for $X^\dagger \times^{\mathrm{fine}}_{S^\dagger} T^\dagger = X^\dagger \times_{S^\dagger} T^\dagger$ respectively $X^\dagger \times^{\mathrm{fs}}_{S^\dagger} T^\dagger = X^\dagger \times_{S^\dagger} T^\dagger$. The simplest case is as follows:

Lemma 1.1. *Let $f\colon X^\dagger \to S^\dagger$ be a morphism of fine log schemes, and let $b\colon T^\dagger \to S^\dagger$ be a strict morphism of fine log schemes. Then, $X^\dagger \times^{\mathrm{fine}}_{S^\dagger} T^\dagger = X^\dagger \times_{S^\dagger} T^\dagger$ as log schemes. The analogous result holds for morphisms of fs log schemes and the fs fiber product.*

Proof. Strictness is stable under base change in $\mathrm{Sch}^{\log}$. Thus, $c\colon Y^\dagger \coloneqq X^\dagger \times_{S^\dagger} T^\dagger \to X^\dagger$ is strict. Thus, for every $y \in Y$, we can find an open neighborhood $h(y) \in X^\circ \subseteq X$ and a strict morphism $h^{-1}(X^\circ)^\dagger \to (X^\circ)^\dagger \to A^\dagger_P$. If $X^\dagger$ is fine, we can choose P to be fine, and then $h^{-1}(X^\circ)^\dagger$ is a fine log scheme. □

Strictness is not necessary for the conclusion to hold. We introduce the following definition.

Definition 1.2. A homomorphism $\theta\colon Q \to P$ of integral monoids is *integral* if, for every homomorphism of integral monoids $\psi\colon Q \to Q'$, the pushout $Q' \oplus_Q P$ in the category CMon is integral. ⧫

Fact 1.3. *The following statements hold:*

(1) *$\theta\colon Q \to P$ is integral if and only if the following holds: For all $q_1, q_2 \in Q$ and $p_1, p_2 \in P$ with $\theta(q_1) + p_1 = \theta(q_2) + p_2$, there are $p' \in P$ and $q_i' \in Q$ with $p_i = \theta(q_i') + p'$ and $q_1 + q_1' = q_2 + q_2'$.*

(2) *Let $\theta\colon Q \to P$ and $\xi\colon P \to R$ be two homomorphisms of integral monoids. If θ and ξ are integral, then $\xi \circ \theta$ is integral. If $\xi \circ \theta$ is integral and ξ is exact, then θ is integral. If $\xi \circ \theta$ is integral and θ is surjective, then ξ is integral.*

(3) *A homomorphism $\theta\colon Q \to P$ between integral monoids is integral if and only if the induced homomorphism $\overline{Q} \to \overline{P}$ is integral.*

(4) *Let $F \subseteq P$ be a face. Then, $P \to P_F$ and $P \to P/F$ are integral.*

(5) *Let $\theta\colon Q \to P$ be integral, and let $F \subseteq P$, $G \subseteq Q$ be faces with $G \subseteq \theta^{-1}(F)$. Then, $Q_G \to P_F$ and $Q/G \to P/F$ are integral.*

(6) *An integral homomorphism $\theta\colon Q \to P$ is exact if and only if $\theta^{-1}(P^\times) = Q^\times$. Such a homomorphism is called* local.

(7) *An integral homomorphism is locally exact.*

(8) *Every $\theta\colon \mathbb{N} \to P$ is integral.*

Example 1.4. (1) $\theta\colon \mathbb{N} \to \mathbb{N}^{\oplus 3}$ with $\theta(1) = (2,2,3)$ is integral.

(2) $\theta\colon \mathbb{N}^{\oplus 2} \to \mathbb{N}$ with $\theta(a,b) = a+b$ is not integral. To see this, consider $q_1 = (1,0)$, $q_2 = (0,1)$, $p_1 = 0$, and $p_2 = 0$. Then, we have $\theta(q_1) = 1 = \theta(q_2)$. Now, if there are $p' \in P$ and $q_i' \in Q$ with $0 = p_i = \theta(q_i') + p'$, then $p' = 0$ and $\theta(q_i') = 0$. But then, $q_i' = 0$, and $(1,0) = q_1 + q_1' \neq q_2 + q_2' = (0,1)$.

(3) We have already seen that $\theta\colon \mathbb{N}^{\oplus 2} \to \mathbb{N}\cdot(1,0) \oplus \mathbb{N}\cdot(-1,1)$ is not integral. Indeed, we considered a pushout of this which is not integral in the previous lecture. ◊

Fact 1.5. *Let $\theta\colon Q \to P$ be a local homomorphism of integral monoids, and assume that Q is sharp. Then, the following statements are equivalent:*

(i) *$\theta\colon Q \to P$ is integral.*

(ii) *$\mathbb{Z}[Q] \to \mathbb{Z}[P]$ is flat.*

(iii) *$\Bbbk[Q] \to \Bbbk[P]$ is flat for some field $\Bbbk$.*

Example 1.6. Consider $\theta\colon \mathbb{N}^{\oplus 2} \to \mathbb{N}\cdot(1,0) \oplus \mathbb{N}\cdot(-1,1)$. This homomorphism is local. We have $\mathbb{C}[x,y] \to \mathbb{C}[z,w]$ given by $x \mapsto z$ and $y \mapsto zw$. The fiber over (x,y) is one-dimensional while all other fibers are zero-dimensional, so $\mathbb{C}[x,y] \to \mathbb{C}[z,w]$ is not flat. ◊

We can do the same for saturatedness:

Definition 1.7. A homomorphism $\theta\colon Q \to P$ between integral and saturated monoids is *saturated* if it is integral, and if for every homomorphism $\psi\colon Q \to Q'$ between integral and saturated monoids, the pushout $Q' \oplus_Q P$ in the category CMon is integral and saturated. ◊

In practice, the most practical saturatedness criterion is often the following. An *ideal* in a monoid is a (possibly empty) subset $K \subseteq P$ with $p + k \in K$ for all $p \in P$ and $k \in K$. An ideal is *radical* if $np \in K$ implies $p \in K$ for $n \in \mathbb{N}_{\geq 1}$.

Fact 1.8. *Let $\theta\colon Q \to P$ be a locally exact (e.g. integral) homomorphism between toric monoids. Then, θ is saturated if and only if $K_\theta = \theta(\theta^{-1}(P \setminus P^\times)) + P$ is a radical ideal in P.*

Example 1.9. (1) $\theta\colon \mathbb{N} \to \mathbb{N}^{\oplus 2}$ given by $\theta(1) = (1,1)$ is saturated.

(2) $\theta\colon \mathbb{N} \to \mathbb{N}^{\oplus 2}$ given by $\theta(1) = (1,2)$ is integral but not saturated. Indeed, we have $(2,2) \in K_\theta$ but $(1,1) \notin K_\theta$. ◊

Saturated homomorphisms have the following nice decomposition.

Fact 1.10. *Let $\theta\colon Q \to P$ be an injective, local, and saturated homomorphism of toric monoids, and assume that Q is sharp. A face $F \subseteq P$ is* θ-critical *if $\theta^{-1}(F) = Q^\times = \{0\}$, and the* θ-critical locus $\mathrm{Crit}(\theta)$ *is the union of the θ-critical faces. Then, we have $P \setminus K_\theta = \mathrm{Crit}(\theta)$, and the summation map*

$$Q \times \mathrm{Crit}(\theta) \to P$$

is bijective.

Example 1.11. Let $P = \mathbb{N} \cdot \langle (1,0,0), (1,1,0), (1,0,1), (1,1,1)\rangle \subseteq \mathbb{Z}^{\oplus 3}$, and let $\theta\colon \mathbb{N} \to P$ be given by $\theta(1) = \bar{t} \coloneqq (1,0,0)$. This homomorphism is integral because the source is $\mathbb{N}$. Let $F_{\bar{x}\bar{u}} = \langle (1,1,0), (1,1,1)\rangle$ and $F_{\bar{y}\bar{u}} = \langle (1,0,1), (1,1,1)\rangle$ be the faces generated by the elements. One may show that $P \setminus K_\theta = F_{\bar{x}\bar{u}} \cup F_{\bar{y}\bar{u}}$ so that K_θ is a radical ideal. Thus, $\theta\colon \mathbb{N} \to P$ is saturated. Now, $\mathbb{N} \times (F_{\bar{x}\bar{u}} \cup F_{\bar{y}\bar{u}}) \to P$ is indeed bijective. ◊

We pass to geometry.

Definition 1.12. We introduce the following notions:

(1) A morphism $f\colon (X, \mathcal{M}_X) \to (S, \mathcal{M}_S)$ between fine log schemes is *integral* if, for every $x \in X$, the induced map $f^\flat\colon \mathcal{M}_{S,f(x)} \to \mathcal{M}_{X,x}$ is an integral homomorphism of monoids.

(2) A morphism $f\colon (X, \mathcal{M}_X) \to (S, \mathcal{M}_S)$ between fs log schemes is *saturated* if, for every $x \in X$, the induced map $f^\flat\colon \mathcal{M}_{S,f(x)} \to \mathcal{M}_{X,x}$ is a saturated homomorphism of monoids. ⧫

Proposition 1.13. *Let $f\colon (X, \mathcal{M}_X) \to (S, \mathcal{M}_S)$ be an integral homomorphism of fine log schemes, and let $b\colon (T, \mathcal{M}_T) \to (S, \mathcal{M}_S)$ be another morphism of fine log schemes. Then, $Y^\dagger \coloneqq X^\dagger \times_{S^\dagger} T^\dagger$ is fine. The analogous result holds for a saturated morphism between fs log schemes.*

Proof. We discuss only the integral case, the saturated case being similar. Let $y \in Y$ be a point, let $x = c(y)$ under $c\colon Y \to X$, and let $s = f(x)$. Let $\beta\colon Q \to \Gamma(S^\circ, \mathcal{M}_S)$ be a chart in an open neighborhood $s \in S^\circ \subseteq S$ which is exact at s, i.e., $\overline{Q} \to \overline{\mathcal{M}}_{S,s}$ is an isomorphism. Then, we can find an open neighborhood $x \in X^\circ \subseteq f^{-1}(S^\circ)$, a chart $\gamma\colon P \to \Gamma(X^\circ, \mathcal{M}_X)$ which is exact at x, and $\theta\colon Q \to P$ compatible with the other maps. Now, we have $\overline{Q} \cong \overline{\mathcal{M}}_{S,s}$ and $\overline{P} \cong \overline{\mathcal{M}}_{X,x}$, so $\overline{Q} \to \overline{P}$ is integral. But then, $\theta\colon Q \to P$ is integral. We conclude by inspecting the proof that $(Y, \mathcal{M}_Y)$ is coherent—the chart constructed there is now fine. □

2 Log derivations

Definition 2.1. Let $f\colon (X, \mathcal{P}_X) \to (S, \mathcal{P}_S)$ be a morphism of pre-log schemes, and let $\mathcal{E}$ be a sheaf of $\mathcal{O}_X$-modules. A *log derivation* is a pair (D, Δ), where $D\colon \mathcal{O}_X \to \mathcal{E}$ is an $f^{-1}\mathcal{O}_S$-linear derivation, and where $\Delta\colon \mathcal{P}_X \to \mathcal{E}$ is a monoid homomorphism with $\Delta(f^\flat(n)) = 0$ for all local sections $n \in \mathcal{P}_S$, and $\alpha_X(m) \cdot \Delta(m) = D(\alpha_X(m))$ for all local sections $m \in \mathcal{P}_X$. We write $\mathrm{Der}^{\log}_{X/S}(\mathcal{E})$ for the sheaf of log derivations. Furthermore, we write $\Theta^{1,\log}_{X/S} \coloneqq \mathrm{Der}^{\log}_{X/S}(\mathcal{O}_X)$ for the *log tangent sheaf*. ◊

$\mathrm{Der}^{\log}_{X/S}(\mathcal{E})$ is a sheaf of $\mathcal{O}_X$-modules. For the addition, we set $(D_1 + D_2)(r) = D_1(r) + D_2(r)$ and $(\Delta_1 + \Delta_2)(m) = \Delta_1(m) + \Delta_2(m)$. For the multiplication, we set $(c \cdot D)(r) = c \cdot D(r)$ and $(c \cdot \Delta)(m) = c \cdot \Delta(m)$. One may check easily that $(D_1, \Delta_1) + (D_2, \Delta_2)$ and $c \cdot (D, \Delta)$ are indeed log derivations.

Remark 2.2. We haven't (yet) established that $\mathrm{Der}^{\log}_{X/S}(\mathcal{E})$ is quasi-coherent. In fact, we did not even assume that $\mathcal{E}$ is quasi-coherent. ◊

When $\phi\colon \mathcal{E} \to \mathcal{F}$ is an $\mathcal{O}_X$-module homomorphism, we obtain an induced map

$$\mathrm{Der}^{\log}_{X/S}(\mathcal{E}) \to \mathrm{Der}^{\log}_{X/S}(\mathcal{F})$$

via $(D, \Delta) \mapsto (\phi \circ D, \phi \circ \Delta)$.

We have a forgetful $\mathcal{O}_X$-module homomorphism

$$\mathrm{Der}^{\log}_{X/S}(\mathcal{E}) \to \mathrm{Der}_{X/S}(\mathcal{E}),$$

and in particular a forgetful homomorphism $\Theta^{1,\log}_{X/S} \to \Theta^1_{X/S}$. This can sometimes be used to compute $\Theta^{1,\log}_{X/S}$.

Fact 2.3. *Let X be an irreducible normal variety over $\operatorname{Spec}\mathbb{C}$, let $E \subset X$ be a reduced effective Weil divisor cut out by an ideal $\mathcal{I} \subset \mathcal{O}_X$, and let $X^\dagger = (X|E)$. Then, we have*

$$\Gamma(X, \Theta^{1,\log}_{X/\mathbb{C}}) \cong \{D\colon \mathcal{O}_X \to \mathcal{O}_X \mid D(\mathcal{I}) \subseteq \mathcal{I}\}$$

for the log derivations over $\operatorname{Spec}(0 \to \mathbb{C})$.

Remark 2.4. I will pose the proof of this as a guided exercise. ◊

Example 2.5. Let $X = \operatorname{Spec}\mathbb{C}[x, y, z]$, and let $E = \{xyz = 0\}$. We have an isomorphism

$$\Theta^1_{X/\mathbb{C}} \xrightarrow{\cong} \mathcal{O}_X^{\oplus 3}, \quad D \mapsto (D(x), D(y), D(z)),$$

for the classical $\mathbb{C}$-linear derivations. Then, we have $D(\mathcal{I}) \subseteq \mathcal{I}$ if and only if $D(xyz) \in (xyz)$, i.e., if and only if $D(x)yz + D(y)xz + D(z)xy \in (xyz)$. This, in turn, is equivalent to $D(x) \in (x)$, $D(y) \in (y)$, and $D(z) \in (z)$. Thus, we have $D = g_x \cdot x\partial_x + g_y \cdot y\partial_y + g_z \cdot z\partial_z$, and $\Gamma(X, \Theta^{1,\log}_{X/\mathbb{C}})$ is a free $\mathbb{C}[x, y, z]$-module of rank 3. ◊

3 Log differential forms

Proposition 3.1. *Let $f\colon (X, \mathcal{P}_X) \to (S, \mathcal{P}_S)$ be a morphism of pre-log schemes. Then, there is a log derivation*

$$(d, \delta)\colon \ (\mathcal{O}_X, \mathcal{P}_X) \to \Omega^{1,\log}_{X/S}$$

with the following universal property: Whenever $(D, \Delta)\colon (\mathcal{O}_X, \mathcal{P}_X) \to \mathcal{E}$ is a log derivation, there is a unique module homomorphism $h\colon \Omega^{1,\log}_{X/S} \to \mathcal{E}$ with $h \circ (d, \delta) = (D, \Delta)$. We say that $\Omega^{1,\log}_{X/S}$ is the sheaf of log differential forms.

Proof. We construct $\Omega^{1,\log}_{X/S}$ as follows. First, we set

$$\Gamma(W, \mathcal{F}) \coloneqq \bigoplus_{g \in \Gamma(W, \mathcal{O}_X)} \Gamma(W, \mathcal{O}_X) \cdot [g] \oplus \bigoplus_{m \in \Gamma(W, \mathcal{P}_X)} \Gamma(W, \mathcal{O}_X) \cdot [\![m]\!].$$

In this $\Gamma(W, \mathcal{O}_X)$-module, we consider the submodule generated by all expressions of the following forms:

$$\begin{aligned}
&[g_1 + g_2] - [g_1] - [g_2] && \text{for } g_1, g_2 \in \Gamma(W, \mathcal{O}_X),\\
&[g_1 g_2] - g_1 \cdot [g_2] - g_2 \cdot [g_1] && \text{for } g_1, g_2 \in \Gamma(W, \mathcal{O}_X),\\
&[f^\sharp(s) \cdot g] - f^\sharp(s) \cdot [g] && \text{for } g \in \Gamma(W, \mathcal{O}_X), s \in \Gamma(W, f^{-1}\mathcal{O}_S),\\
&[\![m_1 m_2]\!] - [\![m_1]\!] - [\![m_2]\!] && \text{for } m_1, m_2 \in \Gamma(W, \mathcal{P}_X),\\
&[\![f^\flat(n)]\!] && \text{for } n \in \Gamma(W, f^{-1}\mathcal{P}_S),\\
&\alpha_X(m) \cdot [\![m]\!] - [\alpha_X(m)] && \text{for } m \in \Gamma(W, \mathcal{P}_X).
\end{aligned}$$

Then, we define $\Gamma(W, \mathcal{Q})$ as the quotient, and $\Omega^{1,\log}_{X/S}$ as the sheafification of the presheaf $\mathcal{Q}$. We set $d(g) = [g]$ and $\delta(m) = [m]$. It is now easy to prove the universal property. □

The universal property of the classical differential forms $\Omega^1_{X/S}$, as well as the construction, gives us a canonical comparison homomorphism $\Omega^1_{X/S} \to \Omega^{1,\log}_{X/S}$.

Let

$$\begin{array}{ccc}
(Y, \mathcal{P}_Y) & \xrightarrow{c} & (X, \mathcal{P}_X)\\
\downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f}\\
(T, \mathcal{P}_T) & \xrightarrow{b} & (S, \mathcal{P}_S)
\end{array}$$

be a commutative diagram of pre-log schemes. Then, we have an induced homomorphism

$$c^* \Omega^{1,\log}_{X/S} \to \Omega^{1,\log}_{Y/T}.$$

Fact 3.2. *In the above diagram, let $(Y, \mathcal{P}_Y) = (X, \mathcal{M}_X)$ be the logification of $(X, \mathcal{P}_X)$, and let $(T, \mathcal{P}_T) = (S, \mathcal{M}_S)$ be the logification of $(S, \mathcal{P}_S)$. Then, $\Omega^{1,\log}_{X/S} \to \Omega^{1,\log}_{Y/T}$ is an isomorphism.*

We can apply the same construction to a morphism of pre-log rings $(Q \to A) \to (P \to R)$ and obtain an R-module $\Omega^{1,\log}_{R/A}$.

Fact 3.3. *Let $\varphi\colon (Q \to A) \to (P \to R)$ be a homomorphism of pre-log rings, and let $f\colon X^\dagger \to S^\dagger$ be the associated morphism of log schemes. Then, $\Omega^{1,\log}_{X/S}$ is the quasi-coherent sheaf associated with the R-module $\Omega^{1,\log}_{R/A}$.*

Example 3.4. Let $(Q \to A) = (0 \to \mathbb{C})$ and $(P \to R) = (\mathbb{N}^{\oplus 3} \to \mathbb{C}[x, y, z])$. Then, $\Omega^{1,\log}_{R/\mathbb{C}}$ is the free $\mathbb{C}[x, y, z]$-module generated by $\frac{dx}{x} \coloneqq [\![x]\!] = \delta(x)$, $\frac{dy}{y} \coloneqq [\![y]\!] = \delta(y)$, and $\frac{dz}{z} \coloneqq [\![z]\!] = \delta(z)$. Since $\frac{1}{x}$ is the derivative of $\log(x)$, this explains the name "logarithmic differential form" and ultimately the name "logarithmic geometry." ◊

Example 3.5. Let P be a toric monoid. Then, we have an isomorphism

$$\mathbb{C}[P] \otimes_{\mathbb{Z}} P^{\mathrm{gp}} \to \Gamma(A_P, \Omega^{1,\log}_{A_P/\mathbb{C}}), \quad 1 \otimes a \mapsto \delta(a).$$

Here, we have extended the log part of the universal derivation to a homomorphism $P^{\mathrm{gp}} \to \mathcal{M}^{\mathrm{gp}}_{A_P} \to \Omega^{1,\log}_{A_P/\mathbb{C}}$. In particular, $\Omega^{1,\log}_{A_P/\mathbb{C}}$ is locally free of rank $\mathrm{rk}(P^{\mathrm{gp}})$. ◊

Corollary 3.6. *Let $f\colon X^\dagger \to S^\dagger$ be a morphism of coherent log schemes. Then, $\Omega^{1,\log}_{X/S}$ is quasi-coherent.*

Proof. A point $x \in X$ admits affine open neighborhoods $f(x) \in S^\circ \subseteq S$ and $x \in X^\circ \subseteq f^{-1}(S^\circ)$ and charts $Q \to \Gamma(S^\circ, \mathcal{M}_S)$ and $P \to \Gamma(X^\circ, \mathcal{M}_X)$ with a compatible homomorphism $\theta\colon Q \to P$. Then, $f\colon (X^\circ)^\dagger \to (S^\circ)^\dagger$ can be identified with $\operatorname{Spec}(P \to \Gamma(X^\circ, \mathcal{O}_X)) \to \operatorname{Spec}(Q \to \Gamma(S^\circ, \mathcal{O}_S))$. □

Remark 3.7. Note that it is not sufficient to assume that $X^\dagger$ and $S^\dagger$ are quasi-coherent, as we do not know the existence of a chart for the morphism in this case. ◊

Example 3.8. Let $X = \operatorname{Spec} \mathbb{C}[t,x,y,u]/(xy - tu)$ and $E = \{t = 0\} \subset X$. Let $X^\dagger = (X|E)$. Then, $\Omega^{1,\log}_{X/\mathbb{C}}$ is *not* quasi-coherent [1, Ex. 1.11]. A full proof of this is a bit of work. In any case, the example shows that log structures without a chart can have unexpected behavior even when arising from seemingly innocent constructions. ◊

Fact 3.9. *The following statements hold:*

(1) *Consider a commutative square*

$$\begin{array}{ccc} Y^\dagger & \xrightarrow{c} & X^\dagger \\ \downarrow{\scriptstyle g} & & \downarrow{\scriptstyle f} \\ T^\dagger & \xrightarrow{b} & S^\dagger \end{array}$$

of log schemes which is Cartesian in $\mathrm{Sch}^{\log}$, $\mathrm{Sch}^{\log}_{\mathrm{fine}}$, *or* $\mathrm{Sch}^{\log}_{\mathrm{fs}}$. *Then, the induced homomorphism* $c^*\Omega^{1,\log}_{X/S} \to \Omega^{1,\log}_{Y/T}$ *is an isomorphism. Furthermore, we have isomorphisms*

$$c^*\Omega^{1,\log}_{X/S} \oplus g^*\Omega^{1,\log}_{T/S} \cong \Omega^{1,\log}_{Y/S} \cong \Omega^{1,\log}_{Y/T} \oplus \Omega^{1,\log}_{Y/X}.$$

(2) *Let* $f\colon X^\dagger \to Y^\dagger$ *and* $g\colon Y^\dagger \to Z^\dagger$ *be two morphisms of log schemes. Then, we have an exact sequence*

$$f^*\Omega^{1,\log}_{Y/Z} \to \Omega^{1,\log}_{X/Z} \to \Omega^{1,\log}_{X/Y} \to 0.$$

(3) *Let* $i\colon Z^\dagger \to X^\dagger$ *be a strict closed immersion, and let* $f\colon X^\dagger \to S^\dagger$ *be a morphism of log schemes. Let* $\mathcal{I}$ *be the ideal sheaf cutting out* $Z \subset X$. *Then, we have an exact sequence*

$$\mathcal{I}/\mathcal{I}^2 \to i^*\Omega^{1,\log}_{X/S} \to \Omega^{1,\log}_{Z/S} \to 0.$$

The left-hand map is given by $h \mapsto i^*dh$.

(4) *Let* $f\colon X^\dagger \to S^\dagger$ *be a strict morphism of log schemes. Then,* $\Omega^1_{X/S} \to \Omega^{1,\log}_{X/S}$ *is an isomorphism.*

Example 3.10. Let $S_0^\dagger = \operatorname{Spec}(\mathbb{N} \to \mathbb{C})$. We have a closed immersion $i\colon S_0^\dagger \to A_{\mathbb{N}}^\dagger$ and furthermore a morphism $f\colon A_{\mathbb{N}}^\dagger \to \operatorname{Spec}(0 \to \mathbb{C})$. Now, $\Omega^{1,\log}_{A_{\mathbb{N}}/\mathbb{C}}$ is the free $\mathbb{C}[t]$-module with basis $\frac{dt}{t} = \delta(t)$. In particular, $i^*\Omega^{1,\log}_{A_{\mathbb{N}}/\mathbb{C}} = \mathbb{C} \cdot \frac{dt}{t}$. The map $\mathcal{I}/\mathcal{I}^2 \to i^*\Omega^{1,\log}_{A_{\mathbb{N}}/\mathbb{C}}$ is given by $t \mapsto dt = t \cdot \frac{dt}{t}$, so it is the zero map. Thus, we have $\Omega^{1,\log}_{S_0/\mathbb{C}} \cong \mathbb{C} \cdot \frac{dt}{t}$. ◊

Example 3.11. Let $X \to \operatorname{Spec} \mathbb{C}$ be an irreducible smooth variety of dimension d, and let $D \subset X$ be a *Zariski normal crossing divisor* in the sense that every $x \in D$ has a Zariski neighborhood $x \in W \subseteq X$ which admits an étale morphism $\phi\colon W \to \mathbb{A}^d =: M$ with $D \cap W = \phi^{-1}(E)$ for some $1 \leq r \leq d$ and $E = \{z_1 \cdot \ldots \cdot z_r = 0\}$. Let $X^\dagger = (X|D)$, and let

$M^\dagger = (M|E)$. Then, $\phi\colon W^\dagger \to M^\dagger$ is strict étale. Thus, we have $\Omega^{1,\log}_{W/M} = \Omega^1_{W/M} = 0$, and hence $\Omega^{1,\log}_{W/\mathbb{C}} = \phi^*\Omega^{1,\log}_{M/\mathbb{C}}$. If $u_i = \phi^\sharp(z_i)$, we may write

$$\Omega^{1,\log}_{W/\mathbb{C}} = \mathcal{O}_W \cdot \frac{du_1}{u_1} \oplus \ldots \oplus \mathcal{O}_W \cdot \frac{du_r}{u_r} \oplus \mathcal{O}_W \cdot du_{r+1} \oplus \ldots \oplus \mathcal{O}_W \cdot du_d.$$

In particular, $\Omega^{1,\log}_{X/\mathbb{C}}$ is locally free of rank d. (In the case where $D \subset X$ is a general normal crossing divisor, we have to allow an étale neighborhood $W \to X$, and then we need to work with log structures in the étale topology.) $\lozenge$

Example 3.12. Let $\phi\colon X \to S = \operatorname{Spec}\mathbb{C}[t]$ be a flat morphism with reduced fibers from an irreducible normal variety X. Let $S^\dagger = (S|0)$, and let $X^\dagger = (X|X_0)$ for $X_0 = \phi^{-1}(0)$. Let $U = (X \setminus X_0) \cup \operatorname{Reg}(X_0)$. Then, $U^\dagger \to S^\dagger$ is strict,[1] so we have an isomorphism $\Omega^1_{U/S} \xrightarrow{\cong} \Omega^{1,\log}_{U/S}$. Since X is normal and $Z = X \setminus U$ has codimension ≥ 2 in X, we have $j_*\mathcal{O}_U = \mathcal{O}_X$ for $j\colon U \to X$ the inclusion. Since

$$\Theta^1_{X/S} \cong \mathcal{H}om(\Omega^1_{X/S}, \mathcal{O}_X) \quad \text{and} \quad \Theta^{1,\log}_{X/S} \cong \mathcal{H}om(\Omega^{1,\log}_{X/S}, \mathcal{O}_X),$$

this implies $\Theta^1_{X/S} \cong j_*\Theta^1_{U/S}$ and $\Theta^{1,\log}_{X/S} \cong j_*\Theta^{1,\log}_{U/S}$. But then, $\Theta^{1,\log}_{X/S} \xrightarrow{\cong} \Theta^1_{X/S}$ is an isomorphism. Note that $\Omega^1_{X/S} \to \Omega^{1,\log}_{X/S}$ is not necessarily an isomorphism at points $x \in Z$. In fact, $\Omega^{1,\log}_{X/S}$ can even fail to be quasi-coherent in this setting, as we have seen in Example 3.8. $\lozenge$

References

[1] Mark Gross and Bernd Siebert. Mirror symmetry via logarithmic degeneration data. II. *J. Algebr. Geom.*, 19(4):679–780, 2010.

[1] Strictness in $\operatorname{Reg}(X_0)$ needs results that we have not discussed.